

INTRODUCTION TO DBMS

Q1) FILL IN THE BLANKS:

1. _____ is collection of interrelated data which helps in efficient retrieval, inserting and deleting of data.
2. A _____ is often abbreviated as DB, is a collection of information organized in such a way that a computer program can quickly select desired pieces of data.
3. A _____ is a software for creating and managing databases.
4. The _____ provides users and programmers with a systematic way to create, retrieve, update and manage data.
5. _____ stores data in such a way that it becomes easier to retrieve, manipulate, and update information.
6. Examples of popular _____ are MySQL, PostgreSQL, Access, Oracle, SQL Server, IBM, DB2 and Sybase.
7. The _____ in a database means only the authorized users can access a database according to its privacy constraints.
8. _____ is designed according to certain rules.
9. The logical structure of a database is called as _____.
10. _____ automatically takes care of backup and recovery.
11. _____ reduces application development and maintenance time.
12. _____ means that the data is accurate and consistent in the database.
13. _____ is very important as there are multiple databases in a DBMS.
14. _____ is vital concept in a database.
15. Only _____ should be allowed to access the database and their identity should be authenticated using a username and password.
16. _____ should not be allowed to access the database under any circumstances as it violates the integrity constraints.
17. _____ Holds a fixed length string (can contain letters, numbers, and special characters).
18. In _____ fixed size is specified in parenthesis.
19. _____ Holds a variable length string (can contain letters, numbers, and special characters).
20. In _____ maximum size is specified in parenthesis.
21. _____ can represent numbers with or without the fractional part.
22. _____ is used for storing integer values.
23. _____ holds the date including day, month and year.
24. _____ holds time. Format: HH:MM: SS.
25. _____ define how a data is connected to each other and how they are processed and stored inside the system.

26. _____describes the method of storing and retrieving the data.
27. The most popular model in DBMS is the _____.
28. _____ is the primary data model, which is used widely around the world for data storage and processing.
29. _____is simple and has all the properties and capabilities required to process data with storage efficiency.
30. Transactions have the following four standard properties, usually referred to by the acronym _____.
31. A _____ is a unit of work that is performed against a database.
32. RDBMS stands for _____.
33. _____ensures that all operations within the work unit are completed successfully.
34. _____ ensures that the database properly changes states upon a successfully committed transaction.
35. _____enables transactions to operate independently of and transparent to each other.
36. _____ensures that the result or effect of a committed transaction persists in case of a system failure.
37. A _____is a basic unit of storage in a relational database.
38. _____consists of rows and columns.
39. A _____ consists of information which is stored under different headings, called as fields or columns.
40. _____are shown vertically in a table. Each field or column has an individual name.
41. A _____is composed of fields and contains all the data about one particular person, company, or item in a database.
42. Record is also called as _____.
43. A _____ or a combination of columns which can be used to identify one or more rows (tuples) in a table is called a key of the table.
44. The group of one or more columns used to uniquely identify each row of a relation is called its _____.
45. _____ is a field (or collection of fields) in one table that refers to the Primary Key in another table.
46. _____link data from individual tables to increase the usefulness of the database.
47. In a _____relationship, one record in a table is associated with one and only one record in another table.
48. _____ can also be viewed as Many-to-One relationships.
49. A _____occurs when multiple records in a table are associated with multiple records in another table.
50. SQL stands for _____

51. _____ is Structured Query Language, which is a computer language for storing, manipulating and retrieving data stored in a relational database.
52. _____ is the standard language for Relational Database Management System.
53. All relational database management systems like MySQL, Base, Oracle, Sybase, Informix, PostgreSQL and SQL Server use _____ as standard database language.
54. _____ became a standard of the American National Standards Institute (ANSI) in 1986, and of the International Organization for Standardization (ISO) in 1987.
58. _____ statements or commands are used to define and modify the database structure of your tables or schema.
59. _____ statements or commands are used to define and modify the database structure of your tables or schema.
60. _____ statements or commands are used for managing data within tables.
61. _____ is used to control user access in a database. It is related to security issue. It is also deals with the rights and permissions of the database access.
62. _____ command creates a new table.
63. _____ command modifies a table.
64. _____ command deletes a table or Database.
65. _____ command Extracts data from a table.
66. _____ command Updates data in a table.
67. _____ command Deletes data from a table.
68. _____ command Insert data into a table.
69. _____ command provide access or privileges on the database objects.
70. _____ command remove access rights or privileges on the database object.

Q.2 STATE TRUE OR FALSE

1. Table is collection of interrelated data which helps in efficient retrieval, inserting and deleting of data. FALSE
2. A database is often abbreviated as DB, is a collection of information organized in such a way that a computer program can quickly select desired pieces of data. TRUE
3. A Database Management System (DBMS) is a software for creating and managing databases. TRUE
4. The MS-Excel provides users and programmers with a systematic way to create, retrieve, update and manage data. FALSE
5. MS-Excel stores data in such a way that it becomes easier to retrieve, manipulate, and update information. FALSE
6. Examples of popular DBMS are MySQL, PostgreSQL, Access, Oracle, SQL Server, IBM, DB2 and Sybase. TRUE
7. The privacy rule in a database means only the authorized users can access a database according to its privacy constraints. TRUE
8. TABLE is designed according to certain rules.
9. The logical structure of a database is called as model. TRUE
10. Database Management System automatically takes care of backup and recovery. TRUE
11. DBMS reduces application development & maintenance time. TRUE

12. Data redundancy means that the data is accurate and consistent in the database. FALSE
13. Data Integrity is very important as there are multiple databases in a DBMS. TRUE
14. Data redundancy is vital concept in a database. FALSE
15. Unauthorised users should be allowed to access the database. FALSE
16. Unauthorised users should not be allowed to access the database under any circumstances as it violates the integrity constraints. TRUE
17. CHAR Holds a fixed length string (can contain letters, numbers, and special characters). TRUE
18. In CHAR fixed size is specified in parenthesis. TRUE
19. VARCHAR Holds a variable length string (can contain letters, numbers, and special characters).
20. In VARCHAR maximum size is specified in parenthesis. FALSE
21. DECIMAL can represent numbers with or without the fractional part. TRUE
22. INT It is used for storing integer values. FALSE
23. DATE holds the date including day, month and year. FALSE
24. TIME () It holds time. Format: HH:MM: SS. TRUE
25. Data models define how a data is connected to each other and how they are processed and stored inside the system. FALSE
26. Data models describes the method of storing and retrieving the data. TRUE
27. The most popular model in DBMS is the Relational model. TRUE
28. Relational data model is the primary data model, which is used widely around the world for data storage and processing. TRUE
29. Relational data model is simple and has all the properties and capabilities required to process data with storage efficiency. FALSE
30. Transactions have the following four standard properties, usually referred to by the acronym ACID. TRUE
31. A transaction is a unit of work that is performed against a database. TRUE
32. RDBMS stands for Relational Database Management System. TRUE
33. Atomicity ensures that all operations within the work unit are completed successfully. TRUE
34. Consistency ensures that the database properly changes states upon a successfully committed transaction. FALSE
35. Isolation enables transactions to operate independently of and transparent to each other. FALSE
36. Durability ensures that the result or effect of a committed transaction persists in case of a system failure. TRUE
37. A table is a basic unit of storage in a relational database. FALSE
38. Table consists of rows and columns. TRUE
39. A table consists of information which is stored under different headings, called as fields or columns. TRUE
40. Columns are shown vertically in a table. Each field or column has an individual name. TRUE
41. A Record is composed of fields and contains all the data about one particular person, company, or item in a database. TRUE
42. Record is also called as Tuple. TRUE

43. A column or a combination of columns which can be used to identify one or more rows (tuples) in a table is called a key of the table.
44. The group of one or more columns used to uniquely identify each row of a relation is called its Primary Key. TRUE
45. Foreign Key is a field (or collection of fields) in one table that refers to the Primary Key in another table. TRUE
46. Relationships link data from individual tables to increase the usefulness of the database. TRUE
47. In a one-to-one relationship, one record in a table is associated with one and only one record in another table. FALSE
48. One-to-Many relationships can also be viewed as Many-to-One relationships. TRUE
49. A many-to-many relationship occurs when multiple records in a table are associated with multiple records in another table.
50. SQL stands for structured query language.
51. SQL is Structured Query Language, which is a computer language for storing, manipulating and retrieving data stored in a relational database. TRUE
52. SQL is the standard language for Relational Database Management System. TRUE
53. All relational database management systems like MySQL, Base, Oracle, Sybase, Informix, PostgreSQL and SQL Server use SQL as standard database language.
54. SQL became a standard of the American National Standards Institute (ANSI) in 1986, and of the International Organization for Standardization (ISO) in 1987. TRUE
58. DDL statements or commands are used to define and modify the database structure of your tables or schema. FALSE
59. DDL statements or commands are used to define and modify the database structure of your tables or schema. TRUE
60. Data Manipulation Language (DML) statements or commands are used for managing data within tables. FALSE
61. DDL is used to control user access in a database. It is related to security issue. It is also deals with the rights and permissions of the database access. FALSE
62. CREATE TABLE command creates a new table. TRUE
63. ALTER TABLE command modifies a table. FALSE
64. DROP TABLE DROP DATABASE command deletes a table or Database. TRUE
65. SELECT command Extracts data from a table. TRUE
66. UPDATE command Updates data in a table. TRUE
67. DELETE command Deletes data from a table.
68. INSERT INTO command Insert data into a table. TRUE
69. GRANT to command provide access or privileges on the database objects. FALSE
70. REVOKE to command remove access rights or privileges on the database object. FALSE

Q2) CHOOSE CORRECT DOUBLE ALTERNATIVES:

- I. A Database Management System is a software for creating and managing databases.
- II. Data model describes the method of storing and retrieving the data.

Q3) CHOOSE CORRECT TRIPPLE ALTERNATIVES:

1. Database is collection of interrelated data which helps in efficient retrieval, inserting and deleting of data.

2. Popular examples of DBMS are MySQL, Access, Oracle, and SQL Server.
3. There are different models like Network model, Hierarchical model & Relational model.

Q2) CHOOSE CORRECT DOUBLE ALTERNATIVES:

- I. A Database Management System is a software for creating and managing databases.
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Q3) CHOOSE CORRECT TRIPPLE ALTERNATIVES:

1. Database is collection of interrelated data which helps in efficient retrieval, inserting and deleting of data.
2. Popular examples of DBMS are MySQL, Access, Oracle, and SQL Server.
3. There are different models like Network model, Hierarchical model & Relational model.

4. SQL is structured query language, which is a computer language for storing, manipulating and retrieving data stored in relational database.

Q4) ANSWER THE FOLLOWING:

A. Distinguish between Data and Information.

Ansa: Data:-

- Data is raw facts
- Data does not help in decision making
- Data could be relevant or irrelevant
- Each students exam score is piece of data.

Information :-

- Information is processed data
- Information helps in decision making
- Without data information cannot be processed
- The average score of a class or a school is information derived from the data.

B. List some applications of DBMS.

- Railway Reservation System
- Library Management System
- Banking
- Universities and Colleges
- Credit Card Transactions

C. Advantages of DBMS:-

Ans. Reducing Data Redundancy : The file based data management systems contained multiple files that were stored in many different locations in a system or even across multiple sytem. This is prevented in a database as there is a single database and any change in it is reflected immediately. Because of this , there is no chance of encountering duplicate data.

Data Intergrity : Data Integrity means that the data is accurate and consistent in the database. Data integrity is important as there are multiple databases in DBMS. All database contain data that are visible to multiple users. So it is necessary that the data is correct and consistent in all the database and for all the users.

Data Security : Data Security is vital concept in a database. Only authorised users should be allowed to access the database and their identity should be authenticated

using a username and password. Unauthorised users should not be allowed to access the database under any circumstances as it violates the integrity constraints.

Sharing of Data : In a database, the users of the database can share the data among themselves. There are various levels of authorisation to access the data, and consequently the data can only be shared based on the correct authorisation protocols being followed.

Privacy : The privacy rule in a database means only the authorized users can access a database according to its privacy constraints. There are levels of database access and a user can only view the data he is allowed to.

D. What are the different properties of transaction? OR Explain the acronym ACID.

Ans : Transactions have the following 4 properties , usually referred by the acronym ACID :-

✓ **Atomicity :** It ensures that all operations within the work unit are completed successfully. Otherwise, the transaction is aborted at the point of failure and all the previous operations are rolled back to their former state.

✓ **Consistency :** It ensures that the database properly changes states upon a successfully committed transaction.

✓ **Isolation :** It enables transactions to operate independently of and transparent to each other.

✓ **Durability :** It ensures that the result or effect of a committed transaction persists in case of a system failure.

E. Define primary key and foreign key.

Ans : **Primary Key :** The group of one or more columns used to uniquely identify each row of a relation is called its Primary Key.

Foreign Key : It is a field (or collection of fields) in one table that refers to the Primary Key in another table.

F. Explain the basic concepts of database.

Ans : The basic concepts of database are :-

📊 **Table :** The table is the basic data storage unit in a Relational database. Table consists of columns and rows.

📊 **Field :** A table consists of information which is stored under different headings, called as fields or columns. Columns are shown vertically in a table.

📊 **Record :** A record is composed of fields and contains all the data about one particular person, company, or item in a database. Record is also called as a Tuple.

■ Key : A column or a combination of columns which can be used to identify one or more rows (tuples) in a table is called a key of the table.

G. Write a short note on different types of relationships.

Ans : One-to-One : In a one-to-one relationship, one record in a table is associated with one and only one record in another table

One-to-Many (or Many-to-One) This is the most common relationship type. In this type of relationship, a row in table City can have many matching rows in table Customer, but a row in table Customer can have only one matching row in table City

Many-to-Many : A many-to-many relationship occurs when multiple records in a table are associated with multiple records in another table. Multiple records in Table Product are linked to multiple records in Table Suppliers.